SPECIFICATIONS & QFD DEVELOPMENT

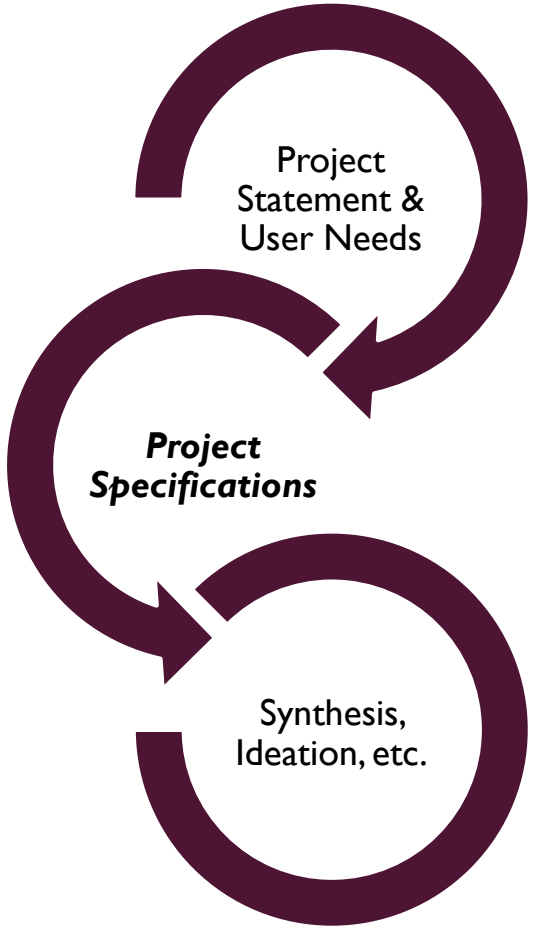
BME362
Week 11
Jennifer Howcroft

TODAY'S TOPICS

- Specifications.
- QFD.

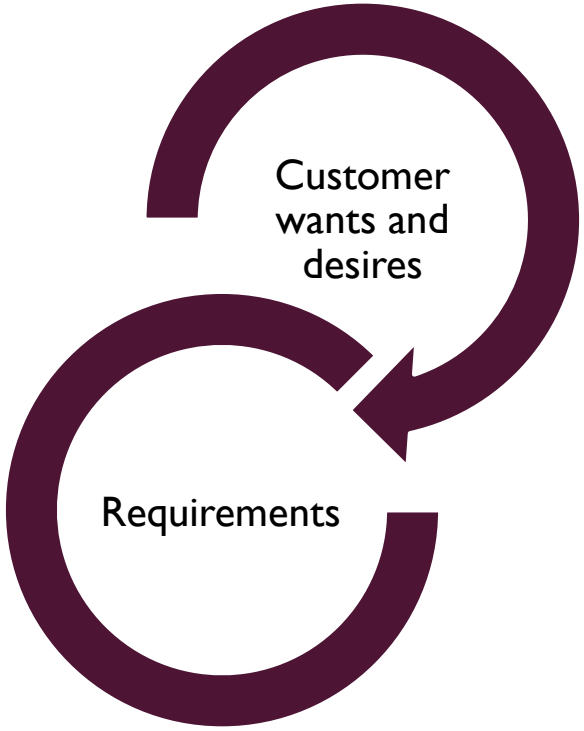
SPECIFICATIONS

ENGINEERING FRAMING TO THE PROBLEM



- Specification:
  - "Document stating requirements" ISO 9000:2015.
  - Includes requirement itself, metric, acceptable values.
- Requirement:
  - "Need or expectation that is stated, generally implied or obligatory" ISO 9000:2015.
  - Physical and performance characteristics expectations.
- Constraint:
  - Limit or restriction on the design space and choices.
  - Intended to prevent product or project failure.

REQUIREMENTS



- Formalizing and detailing statement of need.
- Focus on translating **value** to specifications.
- Design specification that can be optimized for.
- Can be informed from a variety of information sources.

IDENTIFYING REQUIREMENTS

- Informed by your research and problem exploration.

Possible Information Sources

- Stakeholders
- Users
- Clients, Customers
- Standards
- Regulations
- Others

Possible Solution Considerations

- Technical
- Performance
- Manufacturing
- Sales
- Packaging and Transport
- Environmental
- Others

SPECIFYING REQUIREMENTS

Key Elements & Considerations.	What does this mean?
Complete.	Stated in one place. No missing information. No unnecessary information. Provide target metric values: • Marginal: minimal acceptable performance. • Ideal: preferred performance level.
Measurable.	Can test or measure to confirm that the requirement has been met.
Feasible.	Does not violate laws of physics.
Unambiguous.	Clear and concise. Not vague. Not subject to interpretation.
Nonconflicting.	Does not conflict with other requirements or with constraints.
Traceable.	Cite your sources! Standards, regulations, customer provided information, your research, etc.

Modified version of Table 4.1 from Biomedical Engineering Design.

IDENTIFYING CONSTRAINTS

- "Engineering is design with constraints." (S. Florman, Moral Blueprints)

Time	Technical	Financial	Legal and ethical
Regulatory	Industry standards	Social and cultural	Environmental

IDENTIFYING CONSTRAINTS – FOR NOTES

- "Engineering is design with constraints." (S. Florman, Moral Blueprints)
- Time constraints → most common.
  - Often self-imposed.
- Technical design constraints.
  - Impact performance requirements.
  - Conditions in which the product must function (e.g., within the body).
- Financial design constraints.
  - Budgetary and profitability.
- Legal and ethical design constraints.
  - Reduce risk and liability.
  - Patent considerations.
- Regulatory design constraints.
  - Rules that apply to the design of devices, particularly medical devices.
- Industry standards.
  - If meeting an industry standard is a design requirement, then the contents of that standard becomes design constraints.
- Social and cultural constraints.
  - Cultural norms, religion-based restrictions, etc.
- Environmental design constraints.
  - Pollution, hazardous materials, recyclability, landfill disposal, sustainability.

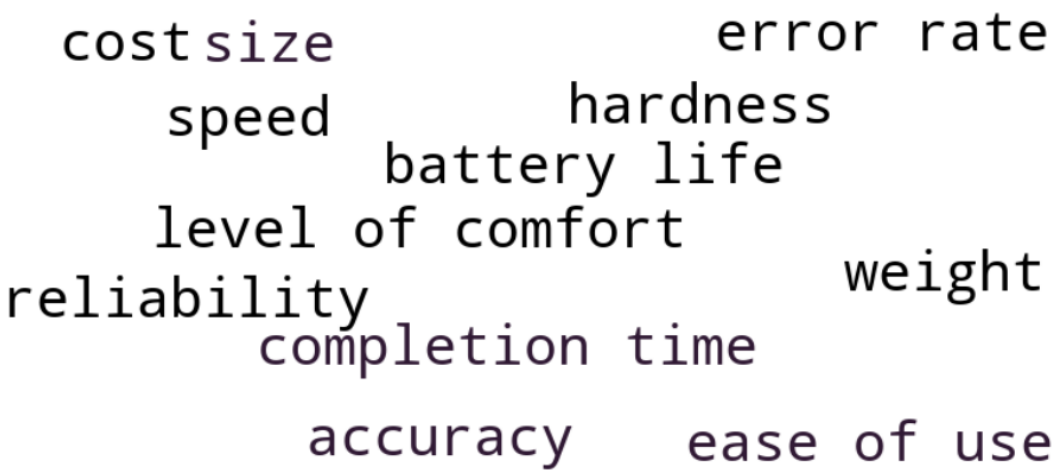
SPECIFYING CONSTRAINTS

Key Elements & Considerations.	What does this mean?
Complete.	Stated in one place. No missing information. No unnecessary information. Provide critical value.
Measurable.	Can test or measure to confirm that the critical value has not been exceeded.
Unambiguous.	Clear and concise. Not vague. Not subject to interpretation.
Nonconflicting.	Does not conflict with other constraints or with requirements.
Traceable.	Cite your sources! Standards, regulations, customer provided information, your research, etc.

Modified version of Table 4.1 from Biomedical Engineering Design.

METRIC

- Means to measure specifications (requirement or constraint).
- Must be appropriate, understandable, and interpretable.
- Should be sensitive to change in the desired scale.
- Should be repeatable.
- Follow industry guidelines and standards.
  - When appropriate and available.

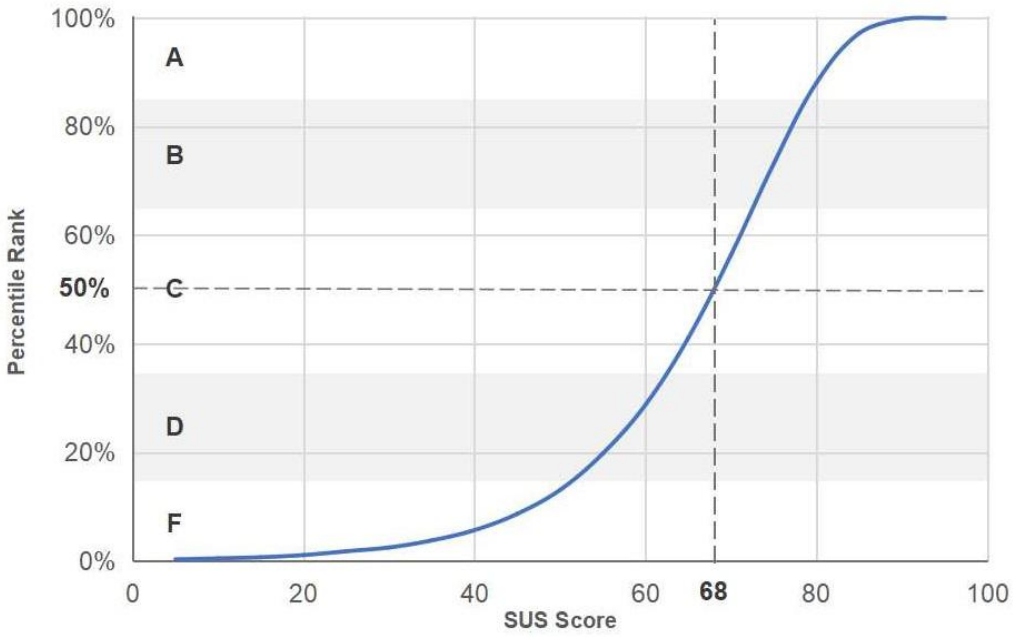


METRIC – TYPES OF VALUE

- Requirement target values:
  - Marginal – Minimally acceptable values.
  - Ideal – Desired performance values.
- Constraint target value:
  - Critical threshold.
- Values should be evidence-backed:
  - Benchmarking against competitor products.
  - Derived from standards and regulations.
  - Informed by problem exploration and research.

EXAMPLE : SYSTEM USABILITY SCALE (SUS)

- System Usability Scale is a widely used measure of overall usability.
- 68% on SUS indicates 'above average' performance.
- Competitor Benchmarks:
  - Excel – 56.5.
  - Word – 76.2.
  - Amazon – 81.8.
  - Google Search – 92.7.



Sauro-Lewis Curved Grading Scale (CGS)

SUS	Grade	Percentile Range
A+	84.1 – 100	96 – 100
A	80.8 – 84.0	90 – 95
A-	78.9 – 80.7	85 – 89
B+	77.2 – 78.8	80 – 84
B	74.1 – 77.1	70 – 79
B-	72.6 – 74.0	65 – 69
C+	71.1 – 72.5	60 – 64
C	65.0 – 71.0	41 - 59
C-	62.7 – 64.9	35 – 40
D	51.7 – 62.6	15 – 34
F	0 – 51.6	0 – 14

https://measuringu.com/interpret-sus-score/

James R. Lewis (2018) The System Usability Scale: Past, Present, and Future, International Journal of Human–Computer Interaction, 34:7, 577-590, DOI: 10.1080/10447318.2018.1455307

EXAMPLE – CATHETERIZATION
(SEE TRANQUILLO ET AL., BIOMEDICAL ENGINEERING DESIGN, PG. 117 – 119)

https://www.besturologistnyc.com/self-intermittent-catheterization-can-it-work-for-you/
https://www.urotoday.com/urinary-catheters-home/intermittent-catheters/description/definition-intermittent-catheters.html
https://www.clinicalcorrelations.org/2019/10/29/intermittent-vs-indwelling-urinary-catheterization-factors-to-consider-for-optimizing-patient-success/

- Key needs (Table 4.2):
  - Safety:
    - Not damage tissue.
    - Not migrate (indwelling only).
    - Low infection rate.
  - Performance
    - Adequate flow rate.
    - Comfortable.
    - Reliable.
    - Function for 30 days (indwelling only).
  - Ease of Use:
    - No assistance required.
    - Easy to insert.
    - Easy to remove.
  - Cost:
    - Affordable.
    - Reimbursable by insurance (region specific).

EXAMPLE – CATHETERIZATION, SELECT REQUIREMENTS (TABLE 4.3, PG. 118)

Description.	Metric.	Ideal Value.	Marginal Value.	Source.
R1. Ease of use. R1.1 Ease of insertion. R1.2 Ease of removal. R1.3 Ease of operation.	 Physician, nurse and patient feedback. Patient feedback. Patient feedback.	 5/5 (5 pt. scale). 5/5 (5 pt. scale). 5/5 (5 pt. scale).	 3/5 (5 pt. scale). 3/5 (5 pt. scale). 3/5 (5 pt. scale).	Nurses and patients.
R2. Adequate flow rate.	Flow rate.	200 mL/min.	100 mL/min.	ASTM standard.
R3. Reasonable cost.	Cost to produce.	$50/month.	$100/month.	Company, benchmarking data.
R4. Comfort.	Patient feedback.	5/5 (5 pt. scale).	3/5 (5 pt. scale).	Physicians, patients.
R5. Low infection rate.	Infection rate.	< 3%.	< 5%.	Physicians, FDA.

EXAMPLE – CATHETERIZATION, SELECT CONSTRAINTS (TABLE 4.3, PG. 118)

Description.	Metric.	Critical Value.	Source.
C1. Will not damage tissue.	Cystoscopic evaluation.	Pass.	Physicians, patients.
C2. Device biocompatibility.	Numerous tests.	Pass.	ISO 10993.
C3. Packaging must maintain sterility.	Accelerated aging and shelf-life stability. Simulated distribution. Package integrity and seal strength	Pass.	ISO 11607.

AS A SENTENCE…

Requirement:

The catheter shall provide an adequate flow rate that is no less than 100 mL/min [1] and ideally 200 mL/min.

[1] ASTM F623-19

Constraint:

The catheter must be biocompatible and conform to all relevant ISO 10993 biocompatibility tests.

The project must not exceed the budget of $50,000.

CLASS ACTIVITY – PROJECT MANAGEMENT CONSTRAINTS

- What are the project management constraints for your project?

Time

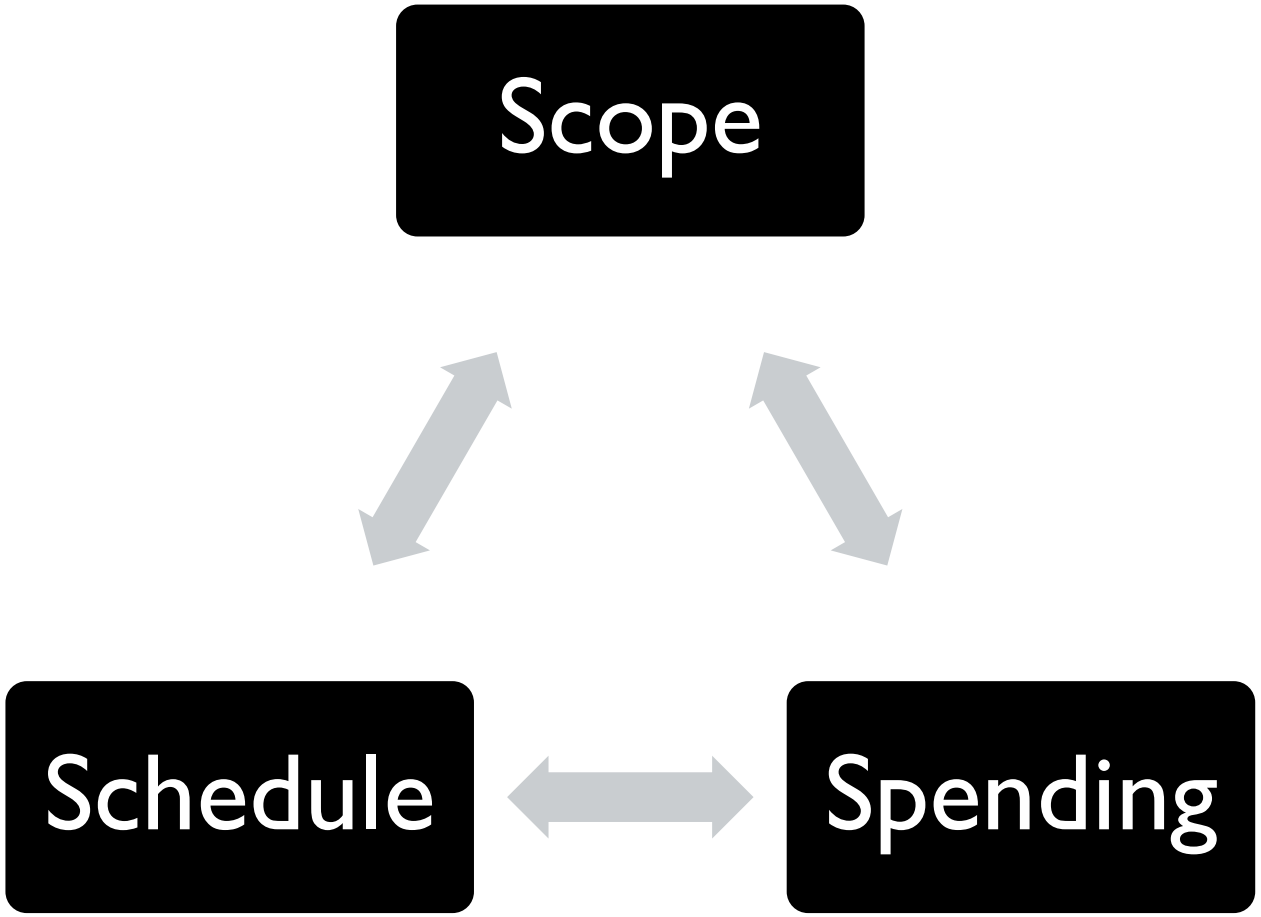
Financial Resources

Non-Financial Resources

Expertise

Deliverables

PROJECT MANAGEMENT – 3 KEY FACTORS



CLASS ACTIVITY – SPECIFICATIONS IN YOUR NEEDS STATEMENT OR SOS

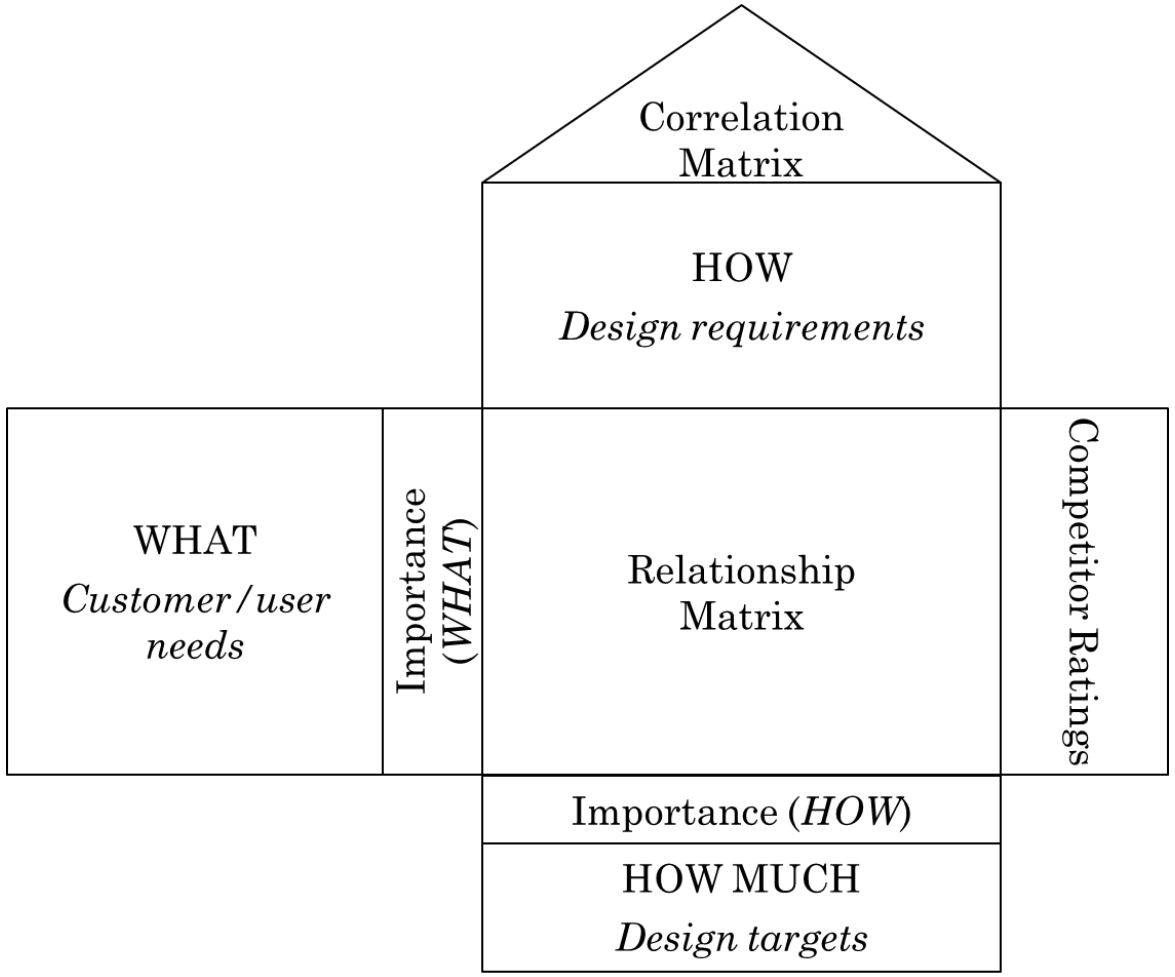
- *"Design a [type of general solution: product, device, or system] to be used by [potential stakeholders/primary users] to [carry out a task or process in an environment] that **[meets a specific set of benchmarks]**."*
- *"Design a [type of general solution: product, device, or system] to address [identified problem] in [population] that **[change in outcome]**."*
- *"A way to address [identified problem] in [population] that **[change in outcome]**."*

- Likely your most important specification.
- Need strong evidence to support this specification:
  - What is the value it represents?
  - What does it mean to optimize?
  - What are the key metrics?
- Discuss with your team.
  - Can you represent this as a complete specification or several specifications?
  - Do you have the needed information?

BREAK

QFD

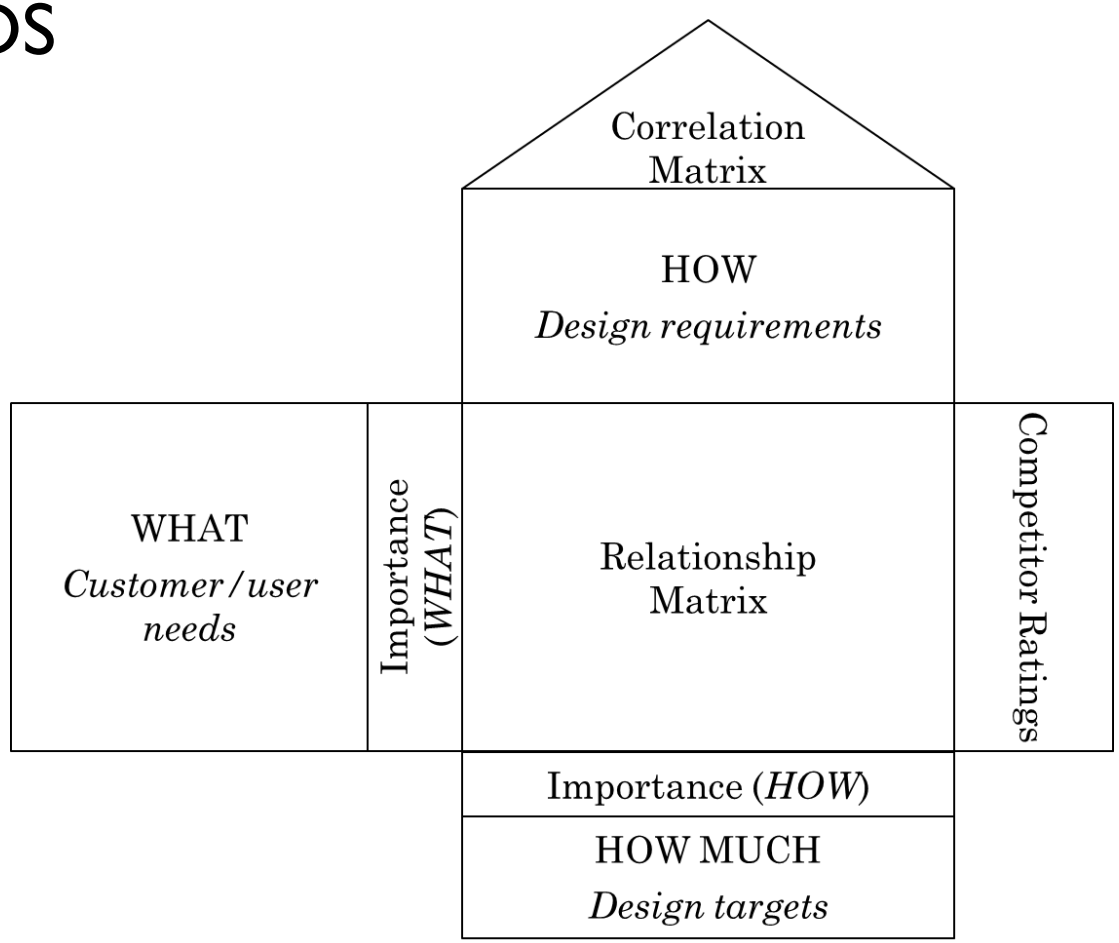
PURPOSE OF QFD



- QFD – Quality Function Deployment chart.
- Also known as a House of Quality.
- Important synthesis tool.
- Organize and analyze information and assessment of project from needs assessment.
- **Compare** customer needs to design requirements.
- **Identify** relationships between customer needs and design requirements and correlations between requirements.
- **Evaluate** competitor products.

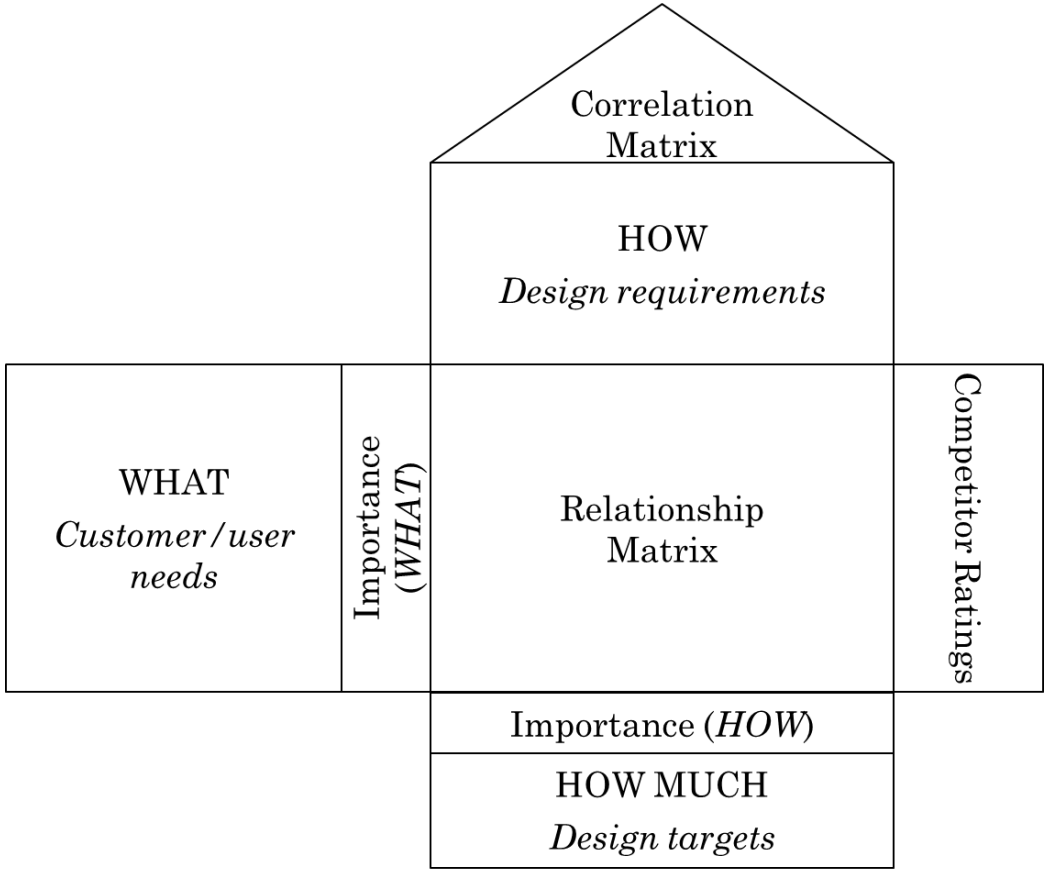
QFD CHART – START WITH USER NEEDS

- WHAT: Customer/user needs:
  - List of the key user needs.
  - Alternate: List basic functions.
- Importance (*WHAT*):
  - Assign relative importance of needs and functions.
    - Based on user insights and other sources of information.
  - These are **weights** not a ranking.
  - Should sum to 100.
  - May be more than one column if you have diverse users with differing priorities for needs.



QFD CHART – THEN FOCUS ON DESIGN REQUIREMENTS

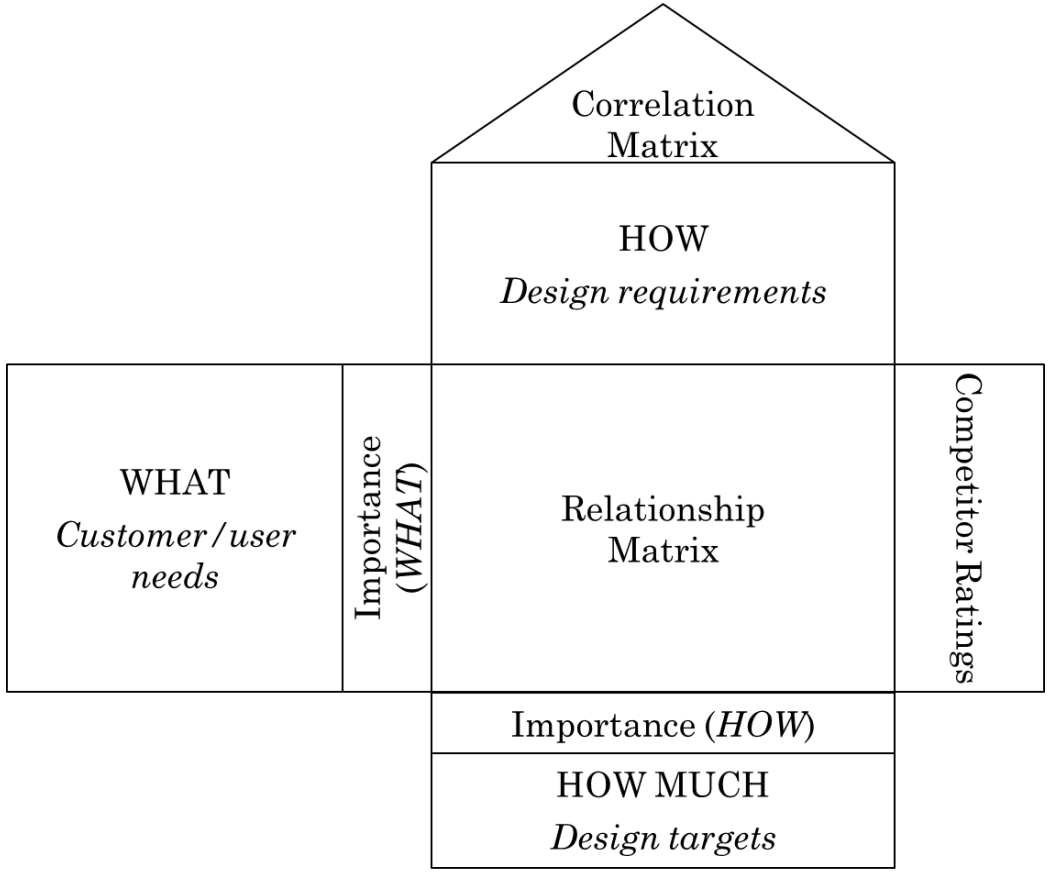
- Engineering Requirements.
  - Focus on technical requirements that will align with user needs.
- Correlation Matrix:
  - Interdependencies between engineering requirements.
  - '--': strong negative, '-': negative, '+': positive, '++': strong positive.
  - May be subjective.
  - Should be justifiable.
  - Identifies synergies and trade-offs.



QFD CHART – EXAMINE RELATIONSHIPS

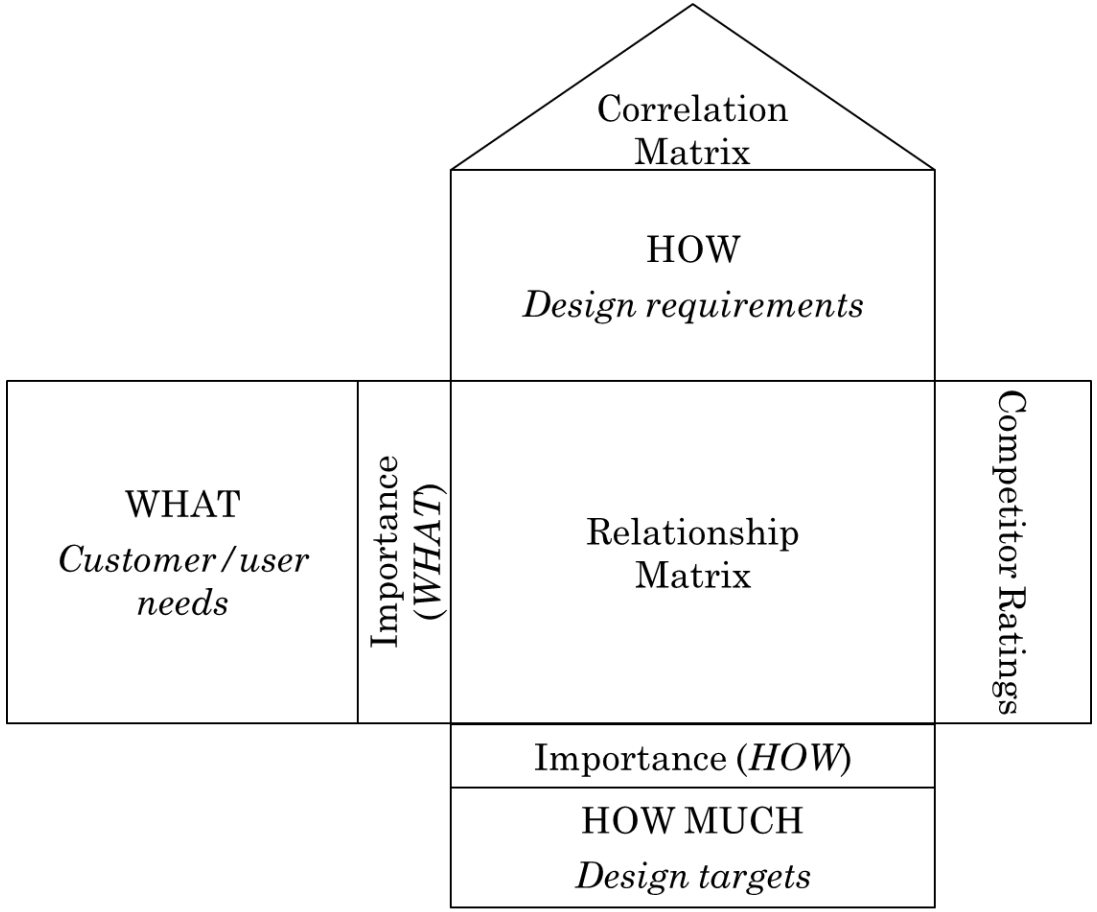
- Relationship Matrix:
  - Interdependencies between user needs and engineering requirements.
  - 0: No interdependency, 1: Slight, 3: Moderate, 9: High.
  - Should be justifiable.

Strength of Relationships Used in the relationship matrix		
●	Strong	9
○	Medium	3
△	Weak	1



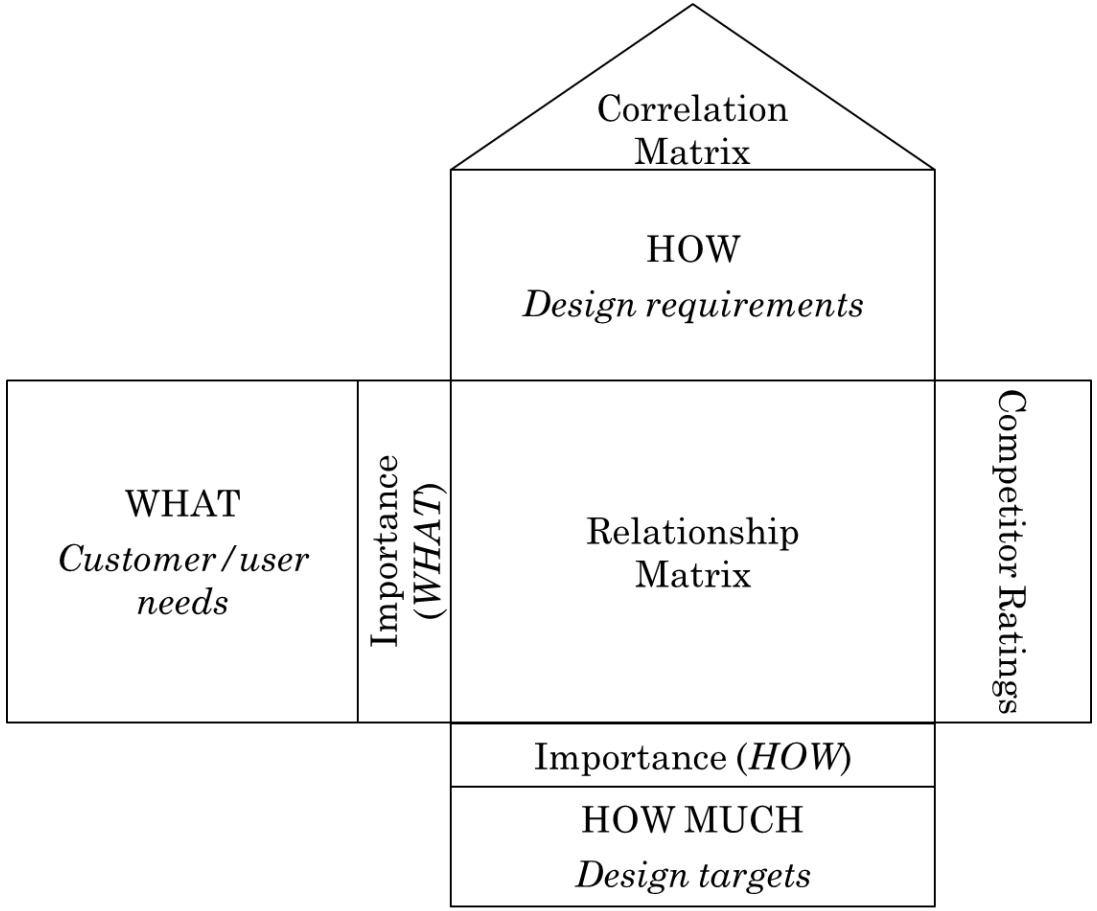
QFD CHART – CONSIDER COMPETITORS

- Competitor Ratings:
  - How well do competitive products meet user needs.
  - 1-Poor to 5-Extremely Well.
  - Possible information sources:
    - Stakeholder insights.
    - Publicly available customer comments.
    - Company information.



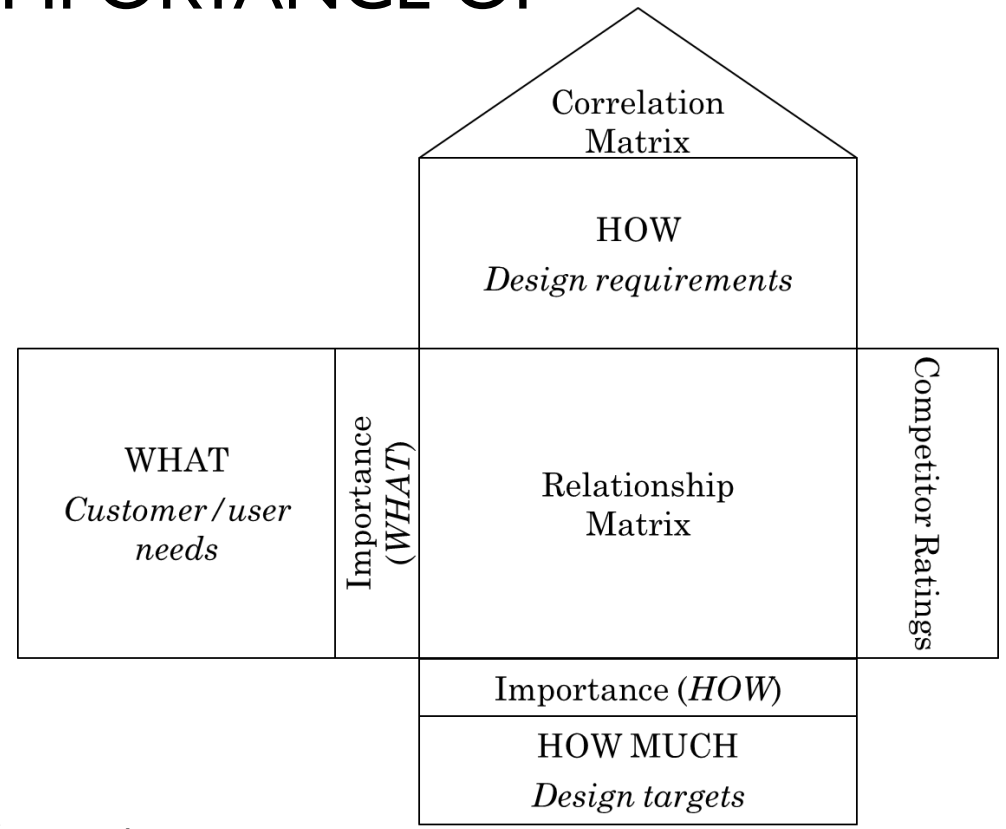
QFD CHART – CAPTURE YOUR METRICS

- HOW MUCH: Design Targets:
  - Marginal – Minimally acceptable values.
  - Ideal – Desired performance values.
  - Include unit of measurement for context.
- Optional – add rows for competitor products.
  - Capture their performance on these key design requirements.



QFD CHART – EVALUATE RELATIVE IMPORTANCE OF REQUIREMENTS

- Importance (*HOW*):
  - Use the Importance (*WHAT*) weightings and the Relationship Matrix to determine the relative importance of the engineering requirements.
  - For each requirement:
    - Multiply importance weighting of each user need with the relationship interdependency (0, 1, 3, or 9).
    - Sum across all user needs.
  - Calculate the % importance of each requirement.
  - If you have multiple columns for importance – what, you will have multiple rows for importance – how.



$$Importance_{Req1} = \sum_{i=1}^{n} \left(Importance_{Need,i} \times Relationship_{Need,iReq1} \right)$$

$$\%\ Importance_{Req1} = \frac{Importance_{Req1}}{\sum_{i=1}^{n} Importance_i} x100\%$$

QFD EXAMPLE – SCENARIO – IF TIME

During the height of the pandemic, one case of COVID-19 in a retirement or nursing home facility was all it took for an outbreak to be declared at that facility. This often led to restricted access to this facility, including the cessation of in-person social visits for residents. This had a negative emotional and mental impact on residents. The reduced interaction also worsened medical conditions like dementia. Due to limited technical skills and technology availability, it can be challenging for these facilities to make technology-based socialization work.

They have approached your design team to tackle this problem.

REFERENCES

- Several design textbooks were used to inform this lecture and create the slide decks. The main sources were:

  - J. Tranquillo, J. Goldberg, R. Allen, "Biomedical Engineering Design", Elsevier, London, UK, 2023.
  - P. Ogrodnik, "Medical Device Design", Academic Press, London, UK, 2020.
  - P.H. King, R.C. Fries, A.T. Johnson, "Design of biomedical devices and systems" 4th Ed., CRC Press, 2018.
  - P.G. Yock, S. Zenios, J. Makower, T.J. Brinton, U.N. Kumar, & F.T. Jay Watkins, "Biodesign: the process of innovating medical technologies" 2nd Ed., Publisher Cambridge University Press.